



IBM DATA SCIENCE PROFESSIONAL CERTIFICATE — CAPSTONE PROJECT

SpaceX Falcon 9 First Stage Landing Prediction

Juan Sebastian escobar

Prepared for the Applied Data Science Capstone review • July 2026

EXECUTIVE SUMMARY

Welcome to the Applied Data Science Capstone Project, where we embark on a journey to predict the successful landing of the Falcon 9 first stage. SpaceX, known for revolutionizing space travel, advertises Falcon 9 rocket launches at a competitive cost of 62 million dollars, compared to other providers' costs upwards of 165 million dollars. This significant cost-saving is largely attributed to SpaceX's ability to reuse the first stage of the rocket. By accurately predicting the landing success of the first stage, we can better estimate the launch costs, providing valuable insights for companies bidding against SpaceX for rocket launches.

METHODOLOGY

The project is structured into several comprehensive modules, each designed to build on the previous one, culminating in a robust predictive model. Below is a detailed description of the tasks and goals achieved in each module.

GITHUB

URL:<https://github.com/JuanSer10/datasciencecoursera/tree/main>

PROJECT GUIDE LINE



Collect

Get and clean DATA
REQUEST to the SpaceX API to gather historical Data



Wrangle

We extract Falcon 9 launch from Wikipedia using Beatifulsoup



Explore

We conducted Extensive EDA Using Matplotlib and Seaborn



Visualize

Use Folium to create Interactive Maps



Predict

Performed hyperparameter tuning using GridSearchCV

METHODOLOGY



Data collection Methodology

Data source from the SpaceX API provided detailed records of Falcon 9 launches



Perform Data Wrangling

Data cleaning involved handling missing values

Standardizing formats

Ensuring consistency



Perform Exploratory Data Analysis

Visualized launch success rates
Payloads

Executed SQL queries to derive insights

Answer specific questions regarding the dataset.

METHODOLOGY



Perform Interactive Visual Analytics

Used Folium to create interactive maps displaying launch sites and outcomes

Developed a plotly Dash application with interactive components like dropdowns



Perform predict Analysis

Built and evaluated various classifications Models , including Logistic Regression

SVN
KNN
Decision Tree

Evaluated models based on accuracy

Data collection – SpaceX API

Step 1 initiate API Request

Use Python requests library to connect to the spaceX API

Endpoint <https://api.spacexdata.com/v4/rockets/>

Step 2 Parse API Response

Convert response from Json to a Python

Extract relevant fields: Launch date, launch site Payload mass Rocket type

Step 3 Store Data Locally

Save extracted data into a pandas DataFrame

Store the DataFrame Locally For Further processing

Github
URL:1
jupyter-labs-
spacex-data-
collection-
api.ipynb.ipynb

Data collection – Wrangling



Step 1 Data cleaning

Identify and fill or remove missing values in the data set
Use techniques to deal with missing data

Step 2 Data Transformation

Convert data types to appropriate formats
Create new features

Normalize / scale numerical

Step 3 Data integration

Merge datasets Collected from different sources
Consists column

Step 4 Data validation

Check duplicate records
Consistency of data entries

GITHUB:

2 labs-jupyter-spacex-Data wrangling.ipynb

DATA WRANGLING

Step 1 Data cleaning

Identify and fill or remove missing values in the dataset
Use appropriate imputation techniques

Step 2 Data Transformation

Convert Data types to appropriate formats
Standardize
Normalize

Step 3 Data Integration

Merge collect from different source
Ensure consistent column names

Step 4 Data Validation

Check for duplicate records
Remove
Verify the accuracy

GITHUB URL:

2 labs-jupyter-
spacex-Data
wrangling.ipynb

EDA WITH SQL

- **Aggregate queries**

Calculated total number of launches

Calculate success rates

Join queries

Joined tables to link Launch records

Combined datasets for comprehensive

- **Filtering Queries**

Filtered data to focus on specific launch

Applied conditions to extract launch

Subqueries

Nested queries to calculate derived metrics

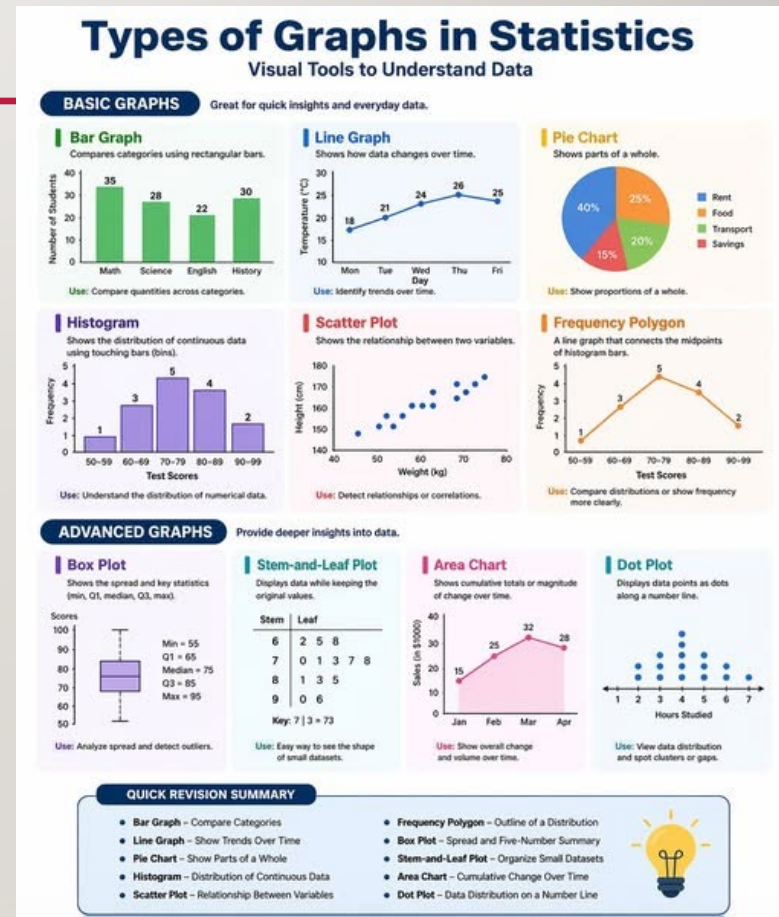
Subqueries used to perform detailed analysis within larger datasets

GITHUB URL: [3 jupyter-labs-eda-sql-coursera_sqlite.ipynb](#)

EDA WITH DATA VISUALIZATION

Exploratory Data Analysis (EDA) is an essential first step in the data science process. It involves analyzing and summarizing a dataset to understand its main characteristics, uncover underlying patterns, spot anomalies, and check assumptions.

GITHUB: [4 jupyter lab dataviz.ipynb](#)



- **Histograms**
- **What they do:** Show the distribution of a single, continuous numerical variable by splitting the data into intervals (called "bins").
- **What they are used for:**

 - Identifying the shape of the data (e.g., normal distribution, skewed left, skewed right).
 - Finding where most values cluster.
 - Spotting outliers or gaps in your data.
 - *Example:* Visualizing the distribution of employee salaries or house prices.
- **2. Bar Charts**
- **What they do:** Compare different categories or groups using rectangular bars, where the length of the bar represents the value.
- **What they are used for:**
 - Comparing discrete categories (not continuous numbers).
 - Tracking changes over time when the changes are large or grouped by year/month.
 - *Example:* Comparing the total sales across different product categories (e.g., Electronics vs. Clothing vs. Groceries).

3. Line Charts

- What they do:** Connect a series of individual data points with a continuous line, typically ordered chronologically.

- What they are used for:**

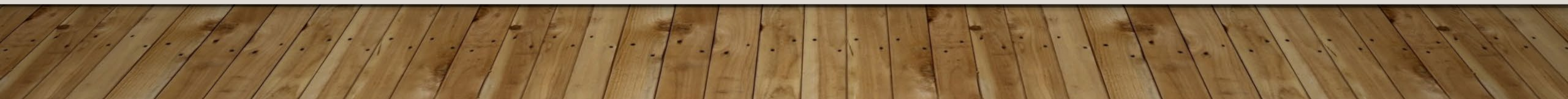
- Tracking trends, cycles, or changes over continuous time intervals.
- Comparing performance or behavior of multiple groups over the same period.
- Example:* Looking at a company's monthly stock price fluctuations over five years.

4. Scatter Plots

- What they do:** Display individual data points on a two-dimensional grid using horizontal (\$X\$) and vertical (\$Y\$) axes.

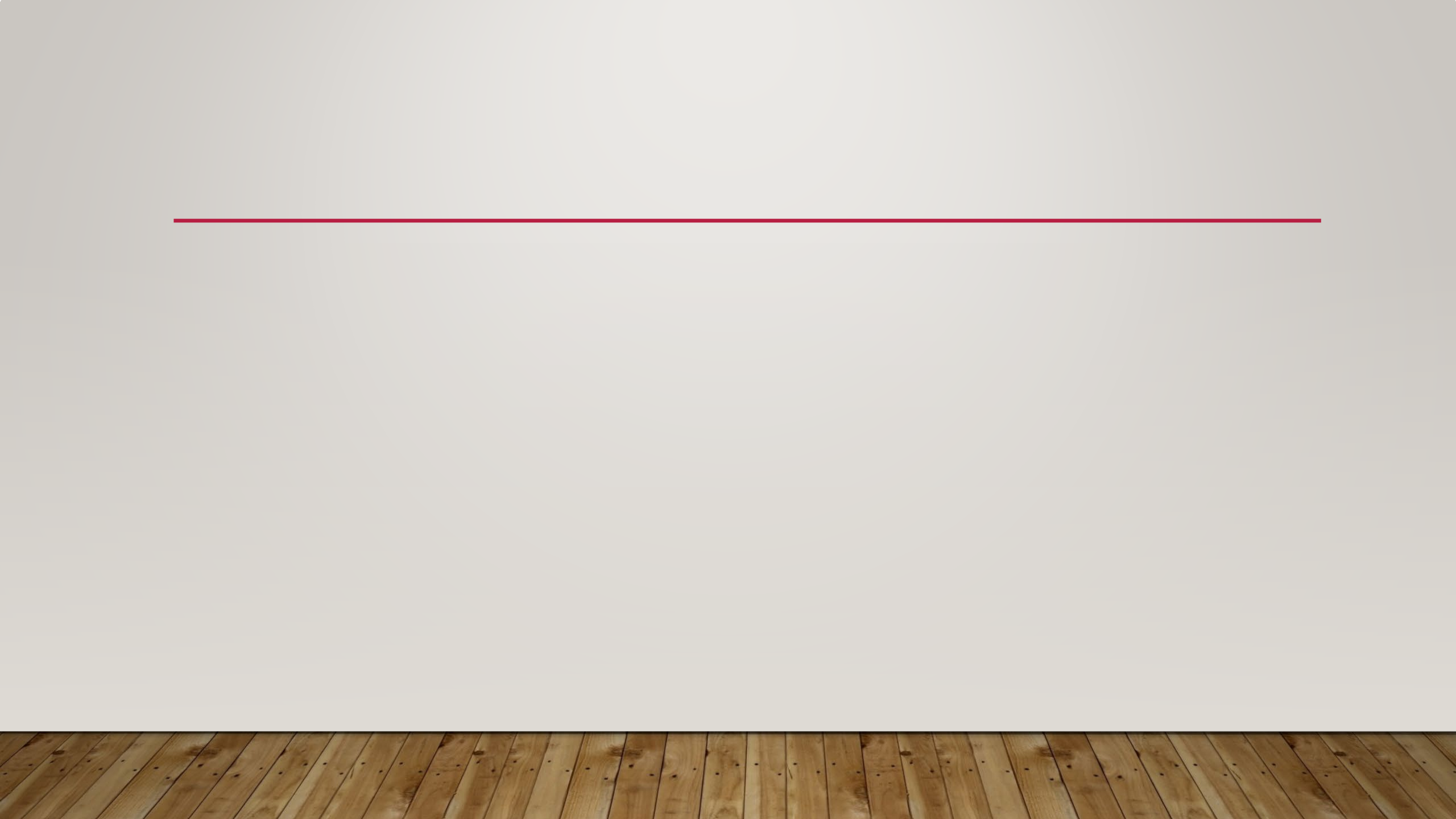
- What they are used for:**

- Determining the relationship or correlation between two continuous variables.
- Checking if an increase in one variable leads to an increase (positive correlation) or decrease (negative correlation) in another.
- Identifying clusters of data or extreme outliers.
- Example:* Plotting a person's height against their weight to see if a correlation exists.



PREDICTIVE ANALYSIS

- **The Selection Process (How it's done)**
- Data scientists rarely just pick one model and hope for the best. They use a systematic process to test and select the winner:
 1. **Establish a Baseline:** Start with a very simple model (like a simple average or basic Linear Regression). Any complex model you build later *must* beat this baseline to be worth the effort.
 2. **Cross-Validation:** Split the historical data into multiple training and testing sets (e.g., K-Fold Cross-Validation). This ensures the model performs well on *unseen* data, preventing **overfitting** (where a model memorizes the training data but fails in the real world).
 3. **Hyperparameter Tuning:** Fine-tune the internal settings of the top-performing models (using techniques like Grid Search or Random Search) to squeeze out the best possible performance.
 4. **Ensemble Methods:** Often, the best model isn't a single algorithm, but a combination of them. Techniques like **Voting** or **Stacking** combine predictions from multiple models to create a stronger, more robust final prediction.
- **GITHUB:6 SpaceX_Machine Learning Prediction_Part_5 (1).ipynb**

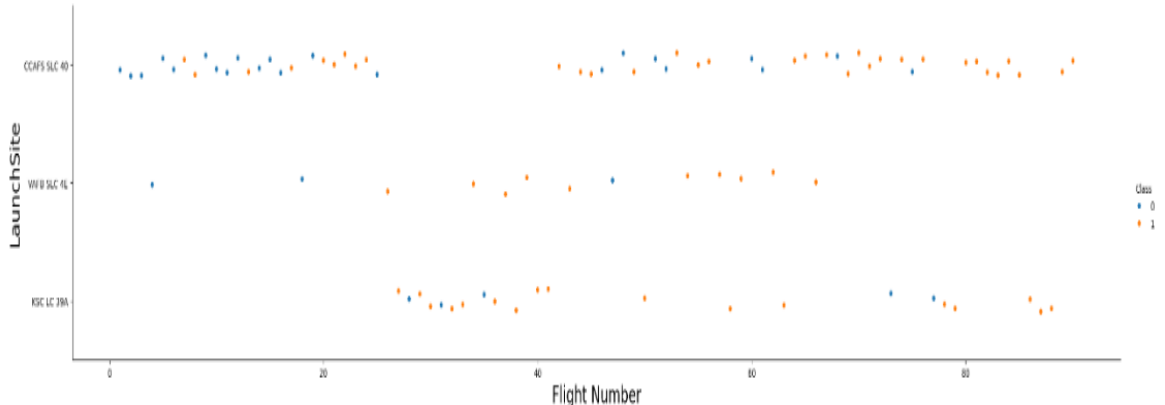


INSIGHTS FROM EDA

DATA
VISUALIZATION

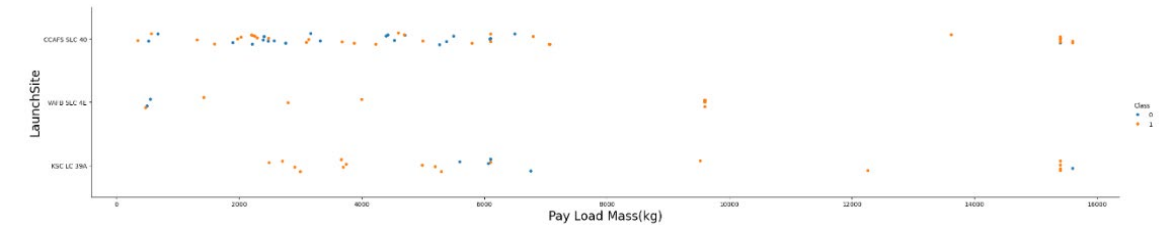


EDA WITH VISUALIZATION · RESULTS (1/2)



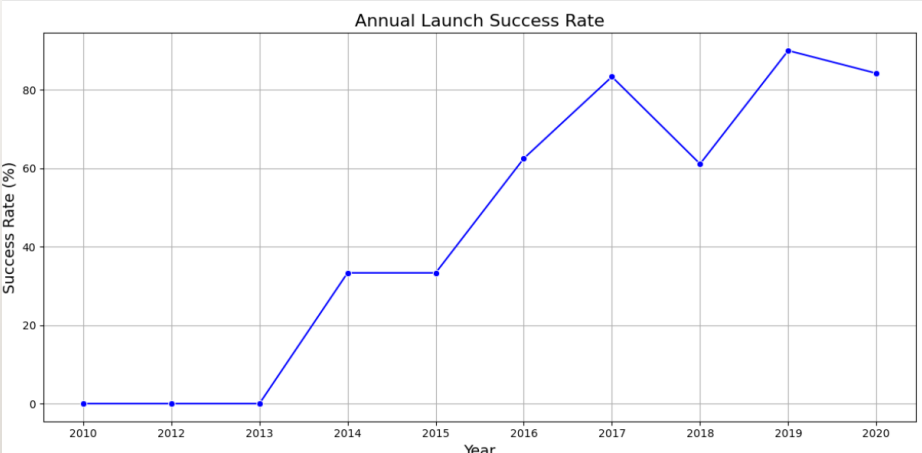
Flight Number vs. Launch Site

Early flights (low FlightNumber) skew toward failed landings (blue); later flights at every site trend orange (success) — experience improves outcomes.



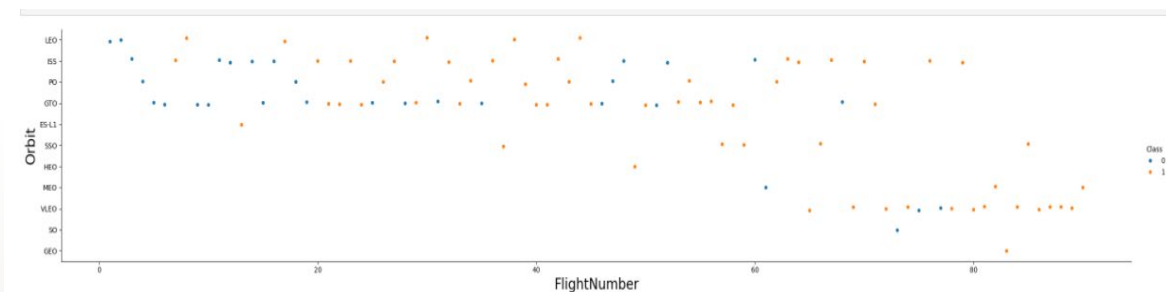
Payload Mass vs. Launch Site

KSC LC-39A handles the heaviest payloads and still lands successfully most of the time; VAFB SLC-4E sees the fewest, lightest launches.



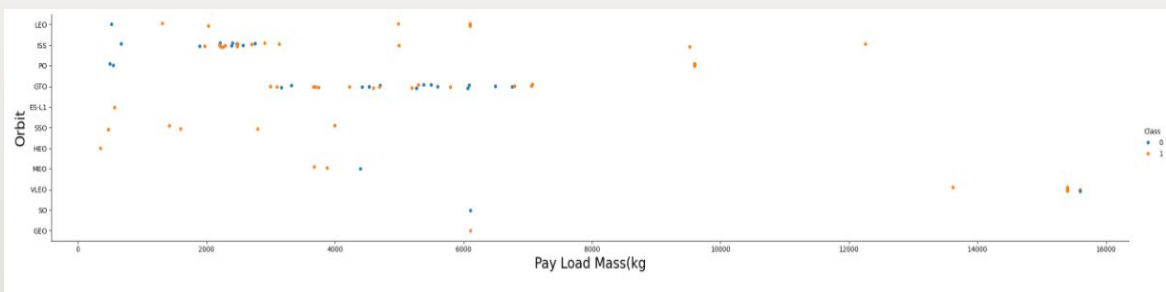
Success rate climbs year over year, from 0% in 2010–2012 to a sustained streak of 100% by 2018 onward.

Orbit type is a strong predictor of success



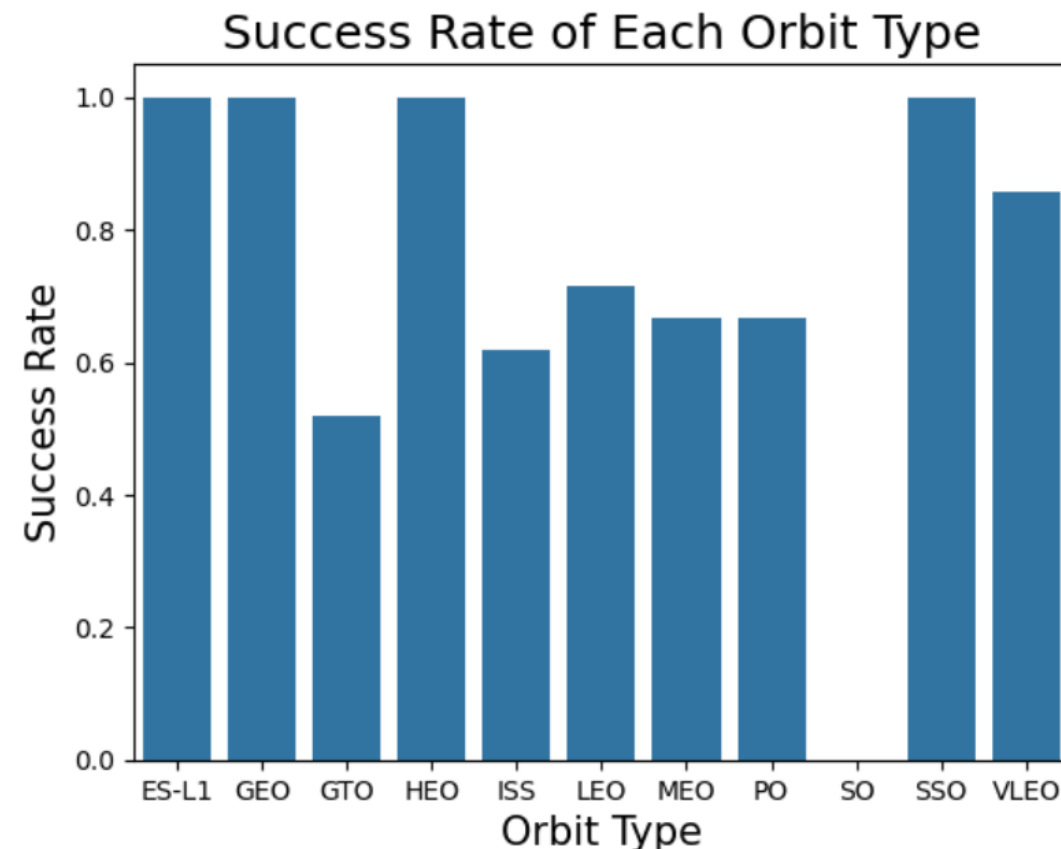
Flight Number vs. Orbit

LEO, ISS, GTO missions span the whole timeline; VLEO missions only appear from flight ~60 onward and land almost every time.



Payload Mass vs. Orbit

GTO missions carry the widest range of payload mass; the heaviest payloads of all (13,000–16,000 kg) go to VLEO and still land.



ES-L1, GEO, HEO and SSO orbits are a perfect 100% in this dataset; GTO is the toughest at ~52%, and SO has zero recorded successes.

LAUNCH SITE NAMES

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT "Launch_site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

TOTAL PAYLOAD MASS

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
] %sql SELECT SUM("PAYLOAD_MASS_KG_") AS "TOTAL_PayloadMass" FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
] TOTAL_PayloadMass
```

45596

LAUNCH SITE NAMES BEGIN WITH CCA

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "launch_Site" LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

AVERAGE PAYLOAD MASS BY F9 V1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql select AVG("PAYLOAD_MASS__KG_") as Avg_payloadMass from SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
```

```
* sqlite:///my_data1.db  
Done.
```

Avg_payloadMass

2928.4

FIRST SUCCESSFUL GROUND LANDING DATE

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql Select MIN(Date) AS "FIRST Successfull landing" From SPACEXTABLE WHERE Landing_Outcome = 'success(ground pad)'
```

```
* sqlite:///my_data1.db  
Done.
```

FIRST Successfull landing

None

SUCCESSFUL DRONE SHIP LANDING WITH PAYLOAD BETWEEN 4000 AND 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE "landing_Outcome"= 'Success(drone ship)' and PAYLOAD_MASS_KG_> 4000 AND PAYLOAD_MASS_KG_< 6000
* sqlite:///my_data1.db
Done.
Booster_Version
```

TOTAL NUMBER OF SUCCESS

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome
```

* sqlite:///my_data1.db
Done.

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

BOOSTER VERSIONS CARRIED THE MAX PAYLOAD MASS

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

LISTS THE RECORDS MONTHS FAILURE LANDING OUTCOME IN DRONE SHIP

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT substr(Date,6,2) AS Month, "Landing_Outcome", Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015' AND "Land:
```

```
* sqlite:///my_data1.db
```

```
Done.
```

	Month	Landing_Outcome	Booster_Version	Launch_Site
	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

RANK THE COUNT OF LANDING OUTCOMES

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_C
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

LAUNCH SITES PROXIMITIES ANALYSIS



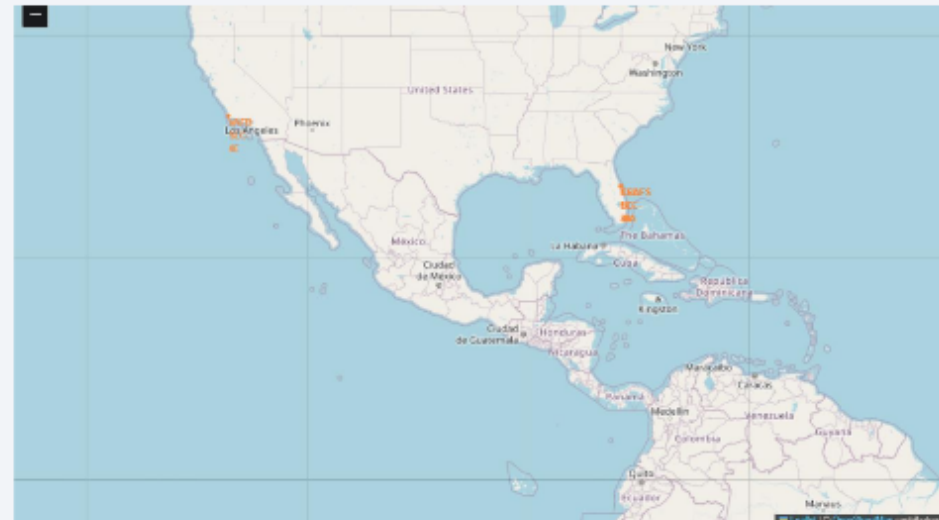
MARK ALL LAUNCH SITES ON A MAP

1. Are all launch sites in proximity to the Equator line?

- No, not all launch sites are in close proximity to the Equator.
- The launch site at Vandenberg Air Force Base (VAFB SLC-4E) is located at a latitude of 34.63, which is further from the Equator compared to the other sites in Florida.

2. Are all launch sites in very close proximity to the coast?

- Yes, all launch sites are in close proximity to the coast.
- The Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.



MARK THE SUCCESS /FAILED LAUNCHES

- This enhanced visualization with clustered markers allows for better exploration and analysis of SpaceX launch data. The clustering makes it easier to manage a large number of markers and observe patterns that might be hidden in a less organized plot. By examining the marker colors and popup information, you can gain deeper insights into the characteristics and distribution of SpaceX launches.
- For example, in the provided screenshot, out of 26 launch sites for CCAFS LC-40, there are 19 red markers and 7 green markers. This color-coding helps to quickly identify the success rate and other categorical distinctions of the launches from this specific site. The red markers might represent unsuccessful launches, while the green markers indicate successful ones, providing immediate visual feedback on the performance of launches at each site.

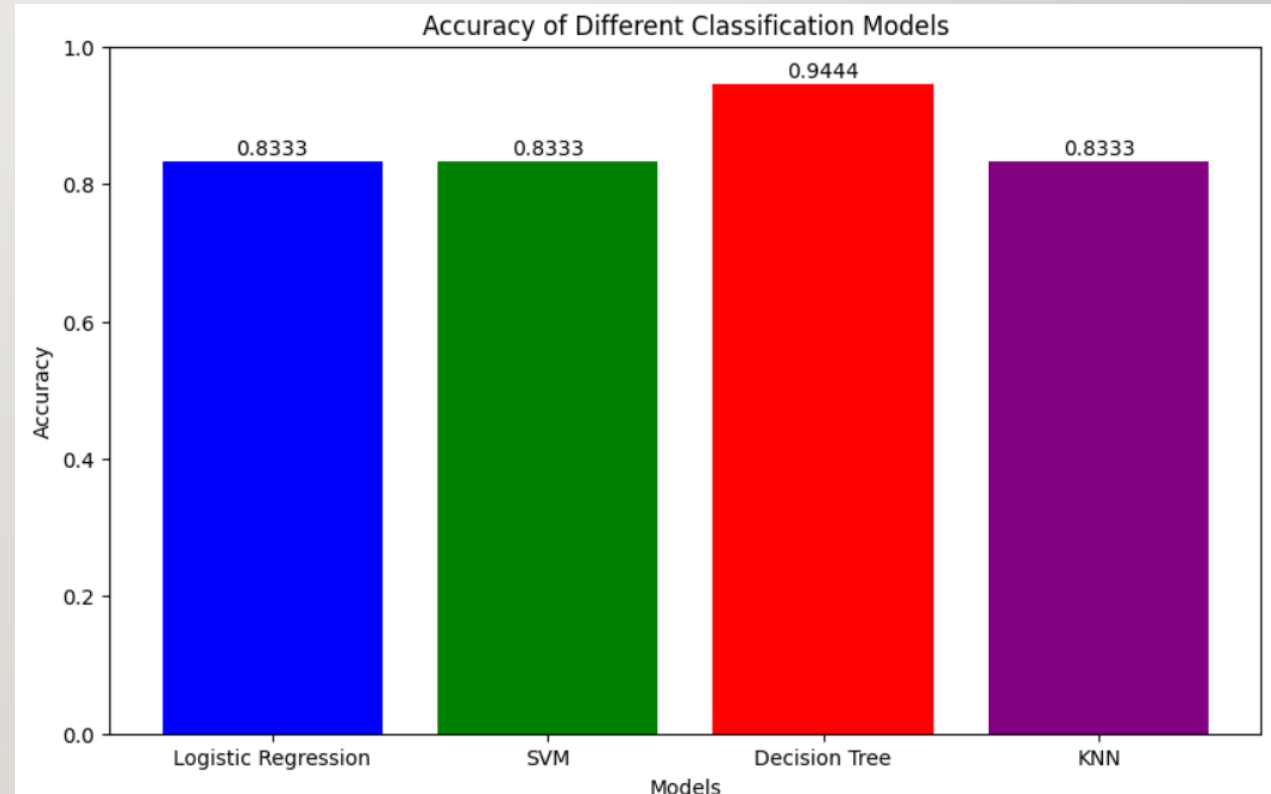


PREDICTIVE ANALYSIS



CLASSIFICATION ACCURACY

Based on the results, the Decision tree is the highest model with an accuracy of 0,9444.



CONCLUSION



-
- Our analysis revealed that the "CCAFS LC-40" launch site has the highest success rate among all sites, accounting for 43.7% of successful launches. This indicates that this site might have optimal conditions or processes that contribute to a higher success rate.
 - The scatter plot analysis showed that the "FT" booster version has a high success rate across various payload masses, demonstrating its reliability and robustness compared to other booster versions. This suggests that future missions might benefit from utilizing this booster version for improved success rates.



Project Repository

All notebooks — data collection, web scraping, data wrangling, EDA with SQL, EDA with visualization, Folium map, Plotly Dash app, and the machine-learning prediction notebook — plus supporting Python files are available at:

<https://github.com/JuanSerl0/datasciencecoursera>

Repository contents:

- `jupyter-labs-spacex-data-collection-api.ipynb`
- `labs-jupyter-spacex-Data_wrangling.ipynb`
- `05_SpaceX_EDA_Data_Visualization.ipynb`
- `spacex-dash-app.py`
- `jupyter-labs-webscraping.ipynb`
- `jupyter-labs-eda-sql-coursera_sqlite.ipynb`
- `06_SpaceX_Interactive_Visual_Analytics_Folium.ipynb`
- `SpaceX-Machine-Learning-Prediction.ipynb`